Feasibility Roadmap for an AlphaFold-Guided Biomolecular Imaging Metasurface

Lachlan Chen
AgInTi Lab, LazyingArt LLC

Auto-curated by AgInTiFlow (https://flow.lazying.art)

Abstract

This report converts the biomolecular imaging metasurface concept into a practical execution roadmap. The conclusion is direct: the project is feasible, but the first working version must be deliberately simple. The first publishable prototype should be a Dps-His6 protein nanocage immobilized on a Ni-NTA surface, carrying or recruiting an optical cargo, with a measured optical signal and a matching structure-to-optics model. This route avoids the main failure mode of the project: combining protein expression, oligomer assembly, surface chemistry, cargo loading, periodic ordering, electromagnetic simulation, and imaging readout before any single layer has been validated. The roadmap gives a stage-by-stage plan, decision gates, failure fixes, required collaborators, and evidence base.

1 Executive Decision

The project should not begin with a complete DNA/protein/RNA metasurface. That version is scientifically exciting, but it combines too many uncertain layers at once. The working route is:

1. Build a biology-first surface-bound nanocage system.
2. Show that it immobilizes reproducibly.
3. Add a strong optical cargo or label.
4. Measure a simple optical signal.
5. Transfer the working biological layer onto a real metasurface or periodic optical chip.

The first prototype is:

Dps-His6 nanocage on a Ni-NTA surface, with optical cargo and optical readout.

This route is realistic because each layer has precedent: Dps/ferritin-like cages are established nanomaterial templates; His-tag/Ni-NTA capture is standard surface chemistry; SpyTag/SpyCatcher offers a covalent backup route; MS2/PP7 gives orthogonal RNA addressing; DNA-origami plasmonics proves that biomolecular scaffolds can position metal nanoparticles; and metasurface biosensing already uses surface-bound biomolecules to shift resonances or enhance fluorescence.

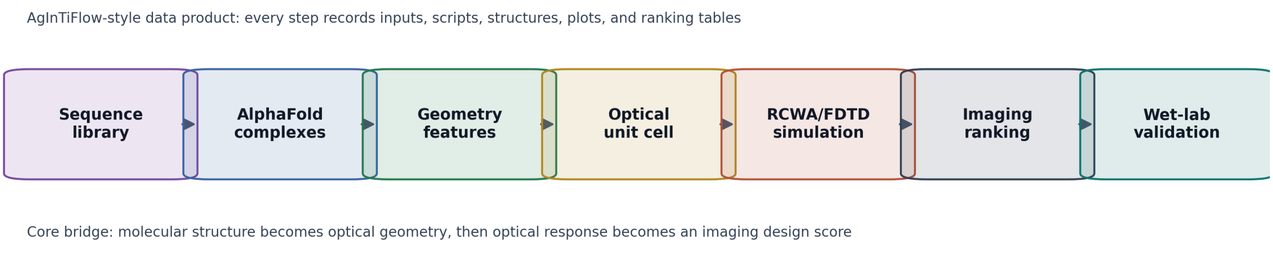


Figure 1: Practical workflow. The project becomes feasible when each layer has a measured output before moving to the next: sequence, structure, geometry, optical abstraction, simulation, ranking, and wet-lab validation.

2 What Is Actually New

The novelty is not that AlphaFold predicts optical spectra. It does not. The novelty is a usable bridge:

biomolecular sequence/complex design → AlphaFold structure prediction → geometry extraction → optical model → experiment → imaging readout.

The strongest defensible claim is:

> **We introduce a structure-guided workflow for converting protein, DNA, and RNA assemblies into optical metasurface design variables, and validate it with a bio-programmed nanocage imaging surface.**

This is a realistic claim. A stronger claim, such as a fully de novo biomolecular metasurface with Nature-level impact, will require a physical prototype with surprising optical performance.

3 Why the Dps-His6 Route Is the First Route

3.1 Biological Feasibility

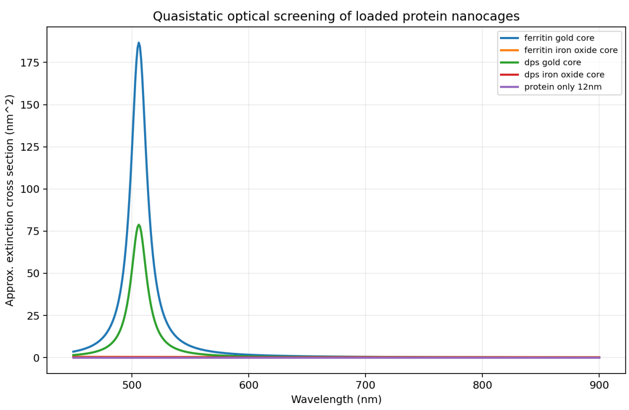
Dps is a compact ferritin-like dodecamer. Its main advantage over ferritin for the first experiment is simplicity: it is smaller, the local AlphaFold screen gives very high confidence for the Dps core, and His6 or SpyTag variants remain structurally strong. Ferritin is still valuable, especially for larger cargo, but it should be a second-generation scaffold.

3.2 Surface Feasibility

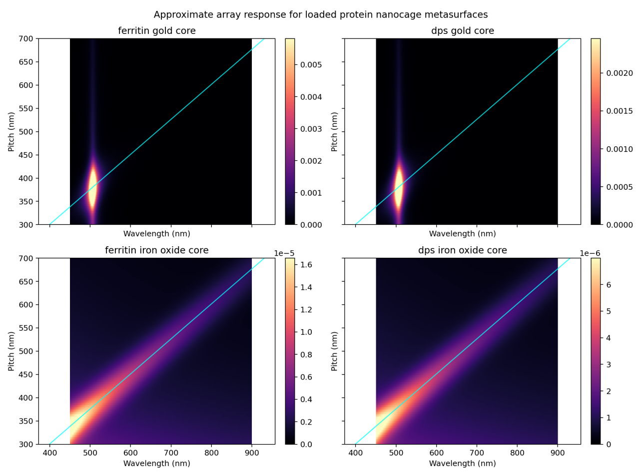
His-tag/Ni-NTA capture is common, cheap, and fast. It is not the strongest possible immobilization chemistry, but it is ideal for the first proof because failure is easy to diagnose. If His6/Ni-NTA is unstable, the project can switch to SpyTag/SpyCatcher or biotin/streptavidin without changing the whole biological concept.

3.3 Optical Feasibility

Protein-only scattering is weak. The protein cage should be treated as a nanoscale carrier and positioning element, not as the main optical material. Optical signal should come from dye, gold, iron oxide, quantum dots, or a resonant substrate.



(a) Cargo makes the signal.



(b) Array pitch gives optical tuning.

Figure 2: First-pass optical screens. The design implication is simple: do not rely on protein-only contrast; add a high-contrast cargo or use a resonant substrate.

4 Feasibility Matrix

Module	Feasibility	Evidence	Main risk	Decision
Dps cage	High	Conserved dodecameric ferritin-like cage; nanotechnology precedent.	Expression or aggregation depends on construct.	First scaffold.
Ferritin cage	High biology, medium first prototype	24-mer nanocage with broad imaging and nanomaterial use.	Larger and slower to optimize.	Second scaffold.
His6/Ni-NTA	High	Standard oriented protein capture on solid surfaces.	Reversible binding and background.	First surface chemistry.
SpyTag/SpyCatcher	High	Spontaneous covalent isopeptide linkage; broad materials use.	Linker can disrupt oligomers.	Backup and modular route.
MS2/PP7	High	Orthogonal RNA hairpin/coat protein systems for RNA imaging.	RNA stability and stoichiometry.	Add after surface proof.
DNA origami plasmonics	High precedent, medium difficulty	Gold nanoparticle helices and chiral plasmonic surfaces are published.	Requires DNA origami and nanoparticle handling.	Stretch path.
Metasurface chip	Medium	Resonance-shift and fluorescence metasurface biosensors exist.	Fabrication and optical alignment.	Add after flat-surface proof.
AlphaFold screening	High as filter	Supports protein/DNA/RNA/ligand/ion complexes.	Does not prove kinetics or surface behavior.	Use for prioritization.

Table 1: Feasibility by layer. The project becomes low-risk only when each layer is validated independently.

5 Local AlphaFold Evidence

The current local AlphaFold campaign supports a practical candidate ranking. Dps, Dps-His6, and Dps-SpyTag are the cleanest core scaffold family. MS2 and PP7 are strong orthogonal RNA adapters. Streptavidin and SpyCatcher/SpyTag are credible connector modules. Ferritin-H-His6 is a useful second-generation scaffold.

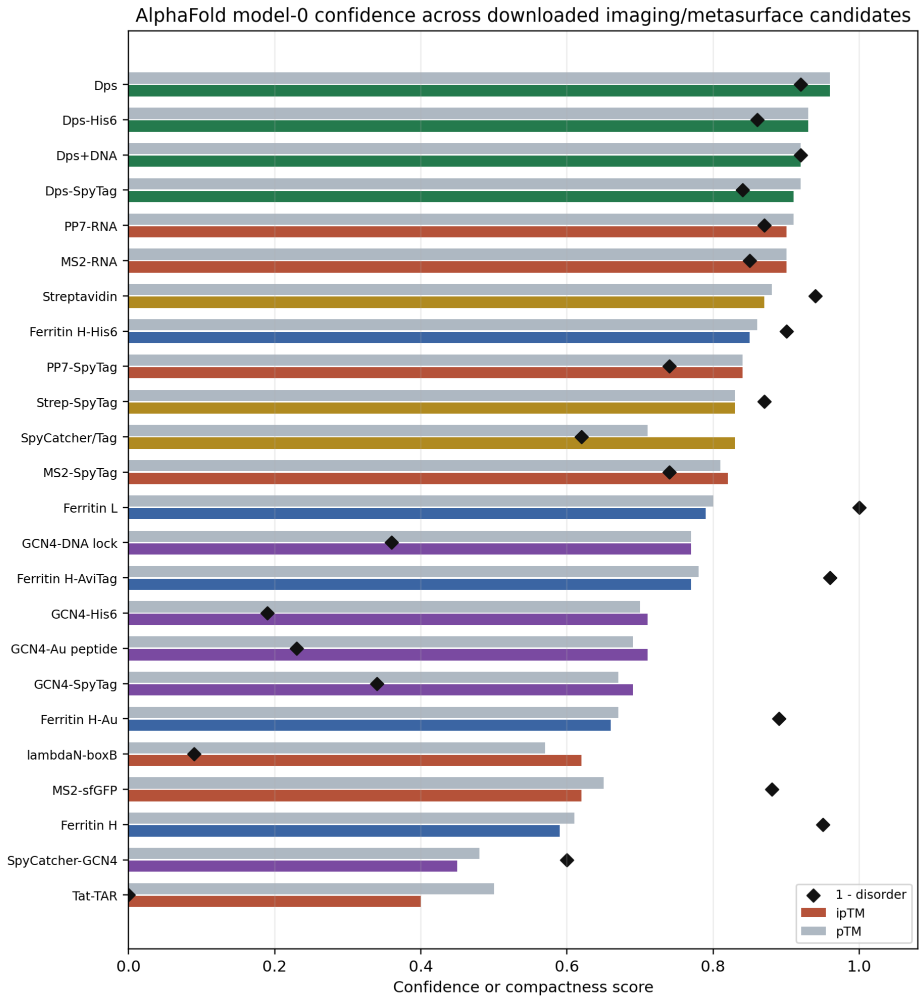


Figure 3: Local model-0 confidence summary for downloaded imaging/metasurface candidates. The first experimental route should prioritize the highest-confidence, easiest-to-build modules rather than the most elaborate architecture.

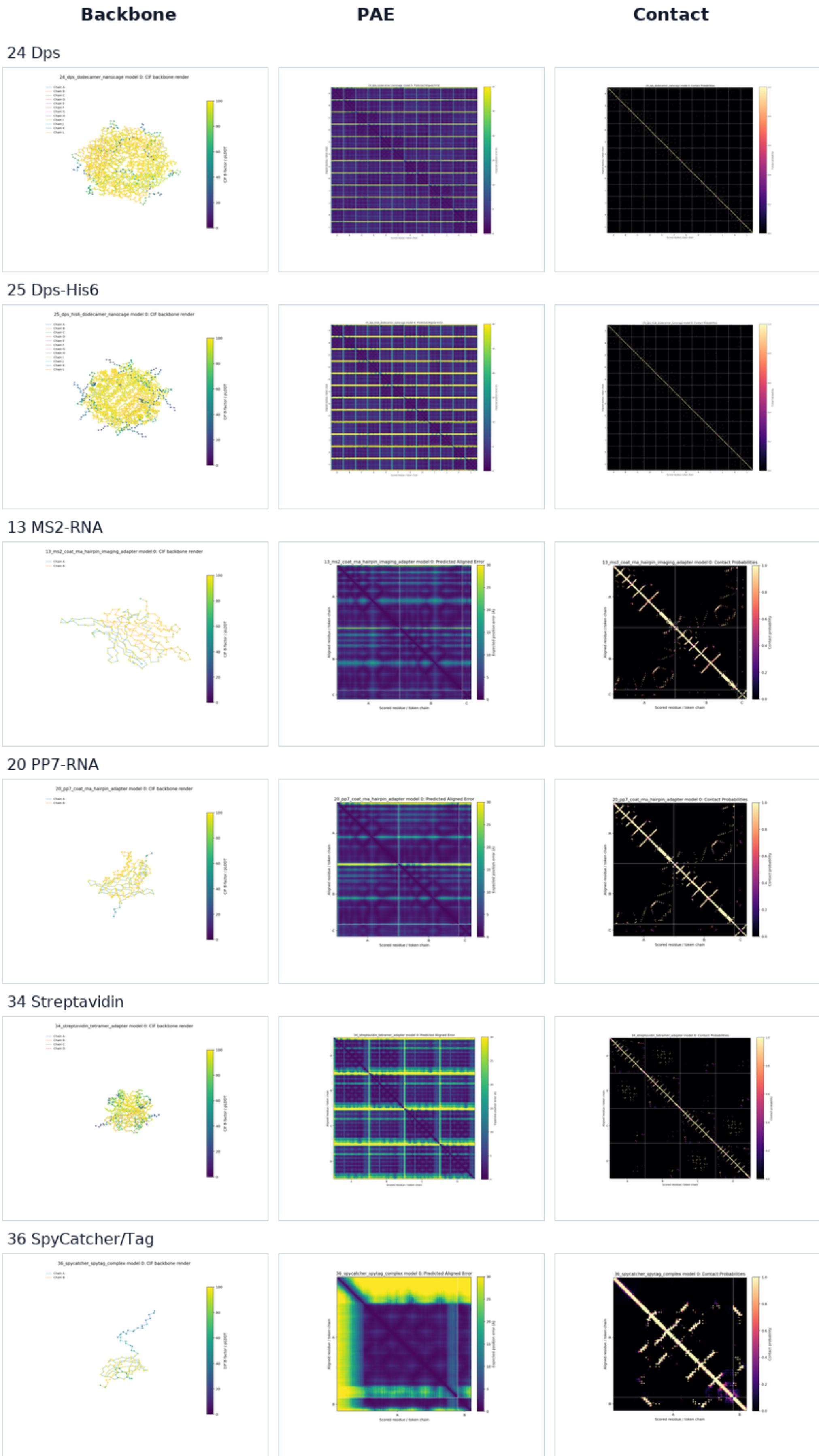


Figure 4: Representative AlphaFold panels for lead candidates. The most actionable result is that Dps-based cages, MS2/PP7 adapters, streptavidin, and SpyCatcher/SpyTag provide a realistic biological toolkit.

6 The Roadmap That Can Really Work

6.1 Stage 0: Freeze the First Version

Goal: avoid endless ideation.

Pick one first architecture:

Dps-His6 nanocage on Ni-NTA surface.

Optional backup:

Dps-SpyTag plus SpyCatcher surface.

Do not include DNA origami, RNA addressing, ferritin, chiral nanorods, and de novo protein design in the first build. Those are later layers.

Deliverables:

- final Dps-His6 sequence;
- optional final Dps-SpyTag sequence;
- AlphaFold model package and geometry table;
- expected oligomer size, handle location, and optical cargo plan.

Pass gate:

- high Dps core confidence;
- low core disorder;
- terminal tag does not visibly destroy dodecamer assembly;
- construct is acceptable to biological collaborators.

6.2 Stage 1: Biology MVP

Goal: prove the protein can be made and remains a cage.

Ask collaborators to express and purify Dps-His6. If resources allow, also prepare Dps-SpyTag and SpyCatcher.

Minimum assays:

- SDS-PAGE for expression and purity;
- SEC or native PAGE for oligomeric state;
- DLS, AFM, negative-stain TEM, or SEC-MALS if available;
- optional mass spectrometry for exact construct confirmation.

Pass criteria:

- soluble protein is obtained;
- purity is good enough for surface work;
- dominant species matches dodecamer-sized Dps;
- DLS PDI is low or microscopy shows reasonably uniform particles;
- no severe precipitation over the experiment window.

Failure fixes:

- move His6 from N terminus to C terminus or vice versa;
- add a short flexible linker such as `GGGGS`;
- shorten or remove unstable terminal handles;
- switch expression host, induction temperature, or solubility tag;
- test unmodified Dps as the assembly control.

6.3 Stage 2: Surface Immobilization MVP

Goal: prove that the cage can be placed on a surface.

Surface options in order of ease:

1. commercial Ni-NTA plate, slide, or biosensor chip;
2. NTA-SAM on gold-coated glass;
3. Ni-NTA functionalized glass;
4. biotin/streptavidin surface if using a biotinylated route;
5. SpyCatcher-coated surface if using Dps-SpyTag.

Minimum assays:

- fluorescence readout if Dps is labeled;
- anti-His or anti-Dps immunostaining;
- AFM on a flat surface;
- QCM-D, SPR, or BLI if available;
- wash stability test.

Pass criteria:

- binding is clearly above surface-only and no-His controls;
- signal remains after washing;
- spatial coverage is reasonably uniform;
- surface density is tunable by protein concentration or incubation time.

Controls:

- surface only;
- Dps without His tag;
- unrelated His-tagged protein;
- imidazole or EDTA competition;
- SpyCatcher-negative control if using SpyTag.

6.4 Stage 3: Add Optical Contrast

Goal: make the biological layer visible to optics.

Recommended options:

1. fluorescently label Dps or a connector protein;
2. attach streptavidin-gold nanoparticles to a biotinylated handle;
3. attach gold nanoparticles through SpyCatcher/SpyTag chemistry;
4. load or mineralize the cage with iron oxide or another core;
5. attach dyes or quantum dots through biotin, maleimide, click chemistry, or Spy chemistry.

Minimum readouts:

- fluorescence microscopy;
- UV-Vis absorption;
- dark-field scattering;
- simple transmission/reflection spectrum;
- optional circular dichroism only after a chiral structure exists.

Pass criteria:

- cargo-loaded surface gives at least 5x signal over protein-only surface;
- signal scales with protein concentration or surface density;
- signal survives washing;
- biological controls remain low.

6.5 Stage 4: Simple Optical Surface Before Real Metasurface

Goal: get a paper-quality optical result without waiting for complex nanofabrication.

Build the first device on Ni-NTA glass, gold-coated NTA-SAM slide, commercial SPR/BLI chip, or a simple patterned substrate.

Measure:

- before/after binding spectrum;
- time-course binding curve if using SPR/BLI/QCM;
- fluorescence or dark-field image;
- surface density estimate;
- protein-only versus cargo-loaded comparison.

Pass criteria:

- reproducible optical change across at least 3 independent samples;
- coefficient of variation below 20–30%;
- controls are clearly separated;
- signal trend is explainable by the optical model.

6.6 Stage 5: Real Metasurface Upgrade

Only after Stage 4 works, transfer the biology to a metasurface.

Best first formats:

1. high-Q dielectric metasurface with a surface resonance;
2. plasmonic nanohole or nanodisk array;
3. periodic gold nanostructure with Ni-NTA or biotin chemistry;
4. silicon or SiN nanodisk array functionalized for protein binding.

Measure resonance shift after Dps binding, resonance shift after cargo loading, fluorescence enhancement, scattering/absorption separation, or polarization response.

Pass criteria:

- resonance shift exceeds measurement noise;
- controls do not produce the same shift;
- cargo-loaded cages give larger or spectrally distinct response;
- optical model matches the trend.

6.7 Stage 6: High-Impact Extension

After the basic surface works, choose one extension:

1. chiral plasmonic pixel: DNA/protein scaffold positions gold nanorods or nanoparticles with left/right handedness;
2. RNA-addressed two-channel pixel: MS2 and PP7 place two labels or cages on one surface;
3. ferritin cargo metasurface: ferritin carries larger mineral or plasmonic cargo;
4. biological state sensor: target binding changes spacing, density, or cargo position.

The chiral plasmonic route is the most exciting. The Dps-His6 route is the safest.

7 Decision Gates

Gate	Question	Pass	If fail
G1	Does Dps-His6 express solubly?	Soluble band and usable purification yield.	Change tag, host, induction, or use unmodified Dps.
G2	Is the cage intact?	SEC/native PAGE/DLS/TEM consistent with dodecamer.	Change tag terminus/linker or remove tag.
G3	Does it bind surface?	Signal above no-His/surface controls.	Optimize Ni-NTA or switch to biotin/SpyTag.
G4	Does binding survive washing?	Retained signal after wash.	Use SpyCatcher/SpyTag covalent route.
G5	Does cargo increase signal?	At least 5x signal over protein-only control.	Switch cargo to gold/dye/QD or increase cargo density.
G6	Is signal reproducible?	n ¿= 3, CV below 20–30%.	Improve surface prep and blocking.
G7	Does model explain trend?	Correct peak, shift, or order of magnitude.	Refine geometry/material assumptions.
G8	Does metasurface improve readout?	Larger shift or enhancement than flat surface.	Choose higher-Q design or stronger cargo.

Table 2: Hard pass/fail gates. These prevent the project from drifting into attractive but unvalidated complexity.

8 Timeline

Table 3: Execution timeline that can realistically produce data.

Time	Goal	Work	Output
Week 1	Freeze constructs	Choose Dps-His6 and optional Dps-SpyTag; review tag location; check AlphaFold geometry.	Final construct table and order list.
Weeks 2–4	Express and purify	Express Dps-His6; purify by Ni-NTA; verify oligomer state.	Purified protein, gel, SEC/native PAGE/DLS/TEM evidence.
Weeks 4–6	Surface binding	Immobilize on Ni-NTA surface; run controls; wash.	Binding image, sensorgram, or surface signal.
Weeks 6–8	Cargo signal	Add fluorescent label, gold, iron oxide, dye, or quantum dot.	Protein-only versus cargo-loaded signal.
Weeks 8–12	Model match	Build optical model from measured geometry and density.	Simulation plot and experiment/model comparison.
Months 3–6	Metasurface transfer	Functionalize a high-Q or plasmonic chip and repeat binding/cargo tests.	First bio-programmed metasurface dataset.
Months 6–12	High-impact extension	Add MS2/PP7, ferritin, chiral plasmonic geometry, or target response.	Stronger paper novelty.

9 Who Does What

9.1 Biology Team

Deliver:

- Dps-His6 construct and optional Dps-SpyTag;
- expression and purification protocol;
- gel images and oligomer-state evidence;
- surface binding and wash-stability evidence;
- cargo attachment or loading evidence;
- raw data with sample concentrations and buffers.

9.2 Optics Team

Deliver:

- flat-surface optical measurement;
- resonance/scattering/fluorescence measurement;
- simple metasurface chip or access route;
- measurement of signal-to-background and reproducibility.

9.3 Computation Team

Deliver:

- AlphaFold and local prediction records;
- geometry extraction from CIF files;
- optical abstraction of cages, cargo, and surface density;
- TORCWA for periodic cells;
- Meep or Tidy3D for 3D particle/cage simulations;
- reports and versioned data tables.

10 What Not To Do First

Avoid these as first experiments:

- full DNA-origami metasurface with multiple gold nanorods;
- de novo designed protein cages;
- live-cell integration;
- many different material-binding peptides at once;
- whole-protein Gaussian calculations;
- final metasurface fabrication before biology binding works;
- claiming top-tier impact before a physical prototype.

These are later-stage ideas. Starting with them will slow the project down.

11 Minimum Tool and Shopping List

11.1 Biology

- Dps-His6 gene or plasmid;
- optional Dps-SpyTag gene or plasmid;
- optional SpyCatcher or SpyCatcher003 plasmid;
- BL21(DE3) or suitable expression host;
- Ni-NTA resin and Ni-NTA surface;
- SEC column or native PAGE setup;
- DLS, AFM, TEM, or access to one;
- fluorescent labeling kit or anti-His/anti-Dps antibody;
- optional streptavidin-gold or biotinylated nanoparticles.

11.2 Optics

- UV-Vis spectrometer;
- fluorescence microscope;
- dark-field microscope for gold nanoparticles;
- simple transmission/reflection setup;
- optional SPR, BLI, or QCM;
- later: high-Q dielectric or plasmonic metasurface chip.

11.3 Computation

- AlphaFold Server for final predictions;
- Chai-1, Protenix, or Boltz-style local model for scalable screening;
- OpenMM or GROMACS for short relaxation if needed;
- TORCWA for periodic arrays;
- Meep or Tidy3D for full 3D simulation;
- Python scripts for geometry, plotting, and reports.

12 Publication Routes

12.1 Short Paper

Claim:

> **AlphaFold-guided selection and experimental validation of protein nanocage surface layers for optical imaging substrates.**

Required figures:

1. workflow diagram;
2. AlphaFold structure screen;
3. Dps expression and assembly validation;
4. surface immobilization;
5. cargo-loaded optical signal;
6. simple simulation matching the trend.

12.2 Stronger Paper

Add:

- real metasurface chip;
- resonance shift or fluorescence enhancement;
- multiple biological modules;
- reproducible imaging field;
- stronger simulation/experiment agreement.

12.3 Top-Tier Paper

Needs at least one surprising result:

- chiral plasmonic bio-metasurface with measured handedness switching;
- multiplexed MS2/PP7 programmable imaging pixels;
- target binding dynamically reconfigures optical response;
- scalable self-assembled bio-optical surface outperforming conventional functionalization.

13 Expected Data Products

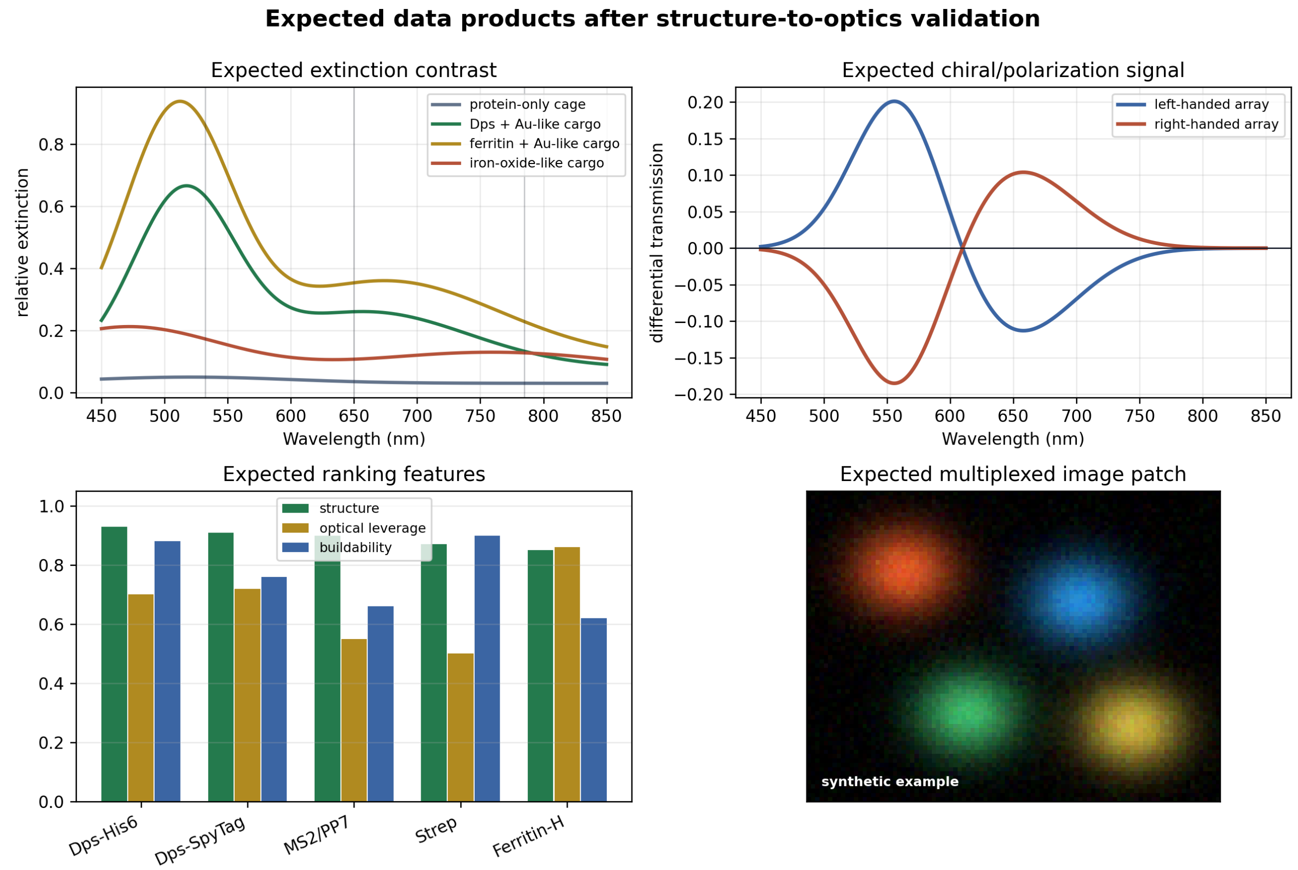


Figure 5: Expected data examples after the first full validation loop. These are synthetic example readouts showing the types of plots needed for a convincing paper: spectra, differential optical response, ranking features, and imaging contrast.

14 Bottom Line

The easiest route with a real chance of working is:

1. Dps-His6.
2. Ni-NTA surface.
3. Fluorescent, gold, iron-oxide, or quantum-dot cargo.
4. Simple optical readout.
5. AlphaFold-derived geometry and optical model.

6. Metasurface transfer after flat-surface success.

This plan is practical because it gives early data even if the final metasurface is delayed. It also creates a clean collaboration boundary: biology proves assembly and surface chemistry; optics proves the imaging signal; computation connects molecular structure to optical design.

15 Evidence Base

References

[1] Dps is a conserved dodecameric ferritin-like DNA-binding protein with nanotechnology relevance. https://pmc.ncbi.nlm.nih.gov/articles/PMC9112962/

[2] Ferritin and Dps cages as supramolecular templates for materials synthesis. https://pmc.ncbi.nlm.nih.gov/articles/PMC3763752/

[3] Ferritin nanocages for imaging, diagnosis, and therapeutic applications. https://www.nature.com/articles/s12276-022-00859-0

[4] Protein cages, rings, and tubes as nanodevice components, including Dps and ferritin size context. https://pmc.ncbi.nlm.nih.gov/articles/PMC3781744/

[5] Protein immobilization on Ni(II)-NTA patterns for oriented His-tag capture. https://research.vu.nl/en/publications/protein-immobilization-on-niii-ion-patterns-prepared-by-microcont/

[6] His-tag/Ni-NTA SAM protocol for QCM and SPR-style measurement. https://www.dojindo.com/products/manual/N475e.pdf

[7] SpyTag/SpyCatcher technology for irreversible protein conjugation and materials. https://pubs.rsc.org/en/content/articlehtml/2020/sc/d0sc01878c

[8] Nature-inspired protein ligation review. https://www.nature.com/articles/s41570-023-00468-z

[9] PP7 coat protein RNA recognition site. https://academic.oup.com/nar/article/30/19/4138/2376092

[10] MS2/PP7 for background-free RNA imaging. https://www.nature.com/articles/srep03615

[11] Structural basis for PP7 and MS2 RNA-protein recognition. https://pmc.ncbi.nlm.nih.gov/articles/PMC3152963/

[12] DNA-based self-assembly of chiral plasmonic nanostructures. https://www.nature.com/articles/nature10889

[13] Surface-bound DNA-origami chiral plasmonic structures with switchable CD spectra. https://www.nature.com/articles/ncomms3948

[14] All-dielectric metasurface biosensing methods. https://pmc.ncbi.nlm.nih.gov/articles/PMC11770831/

[15] Dielectric metasurfaces for label-free sensing, spectroscopy, and chiral sensing. https://pubs.acs.org/doi/10.1021/acsphotonics.0c01030

[16] AlphaFold 3 biomolecular interaction prediction. https://www.nature.com/articles/s41586-024-07487-w

[17] TORCWA GPU-accelerated RCWA and gradient-based metasurface design. https://www.sciencedirect.com/science/article/pii/S0010465522002715

[18] Meep open-source FDTD electromagnetic simulation. https://meep.readthedocs.io/